

Anand Model: Viscoelastoplas ticity and its Application to

Source Paper

Constitutive Equations for Hot-Working of Metals

Author: Lallit Anand
(1985)

DOI: 10.1016/0749-
6419(85)90004-X



Anand 1985 Paper

Case Study: Wang (2001)

Apply to Solder



Wang Paper

Source: Wang, C.
H. (2001). "A
Unified Creep–
Plasticity Model for
Solder Alloys."

DOI:

10.1115/1.1371781

Why Wang's Paper Matters

- Applies Anand's unified viscoplastic framework to model solder behavior.
- Anand's model can

Comparing Anand Model Predictions at Two Strain Rates

Observed Behavior

- **Top Graph (a):**
 $\dot{\epsilon} = 10^{-2} \text{ s}^{-1}$
- High strain

Key Insights from Wang (2001)

- “At lower strain rates, recovery

Main Equations of Wang's Anand-Type Viscoplastic Model

Flow Rule (Plastic Strain Rate)

- $$\dot{\varepsilon}^p = A \exp\left(-\frac{Q}{RT}\right) \left[\sinh\left(\frac{j\sigma}{s}\right) \right]^{1/m}$$

- Plastic strain rate increases with stress

Anand Viscoplasticity Constants for 60Sn40Pb

Image Reference

Values are from
correspond to
60Sn40Pb solder
parameters used in
Anand's model:

- S_0 : Initial deformation resistance

Numerical Values

- $S_0 = 5.633 \times 10^8$ Pa
- $Q/R = 10830$ K
- $A = 1.49 \times 10^{-5}$ s⁻¹
- $\xi = 11$
- $m = 0.303$
- $h_0 = 2.6408 \times 10^8$ Pa

Forward Euler Explicit time integration scheme Pseudocode

Initialization

- Material constants:
 $A, Q/R, j, m, h_0, \hat{s}, n, a, E$
- Strain rate: $\dot{\epsilon}$
- Temperature set: $\{T_i\}$
- Set: $\epsilon^p(0) = 0 \quad \epsilon(0) = \hat{\epsilon}$

Plastic

6. $s^* = \hat{s}$

7. $\dot{s} = h_0$

Forward Euler Scheme for Anand Model

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

# Material constants for 62Sn36Pb2Ag solder alloy
A = 2.24e8          # 1/s
Q_R = 11200        # K
j = 13             # dimensionless
m = 0.21           # dimensionless
h0 = 1.62e10       # Pa
s0 = 8.47e7        # Pa
s_hat = 8.47e7     # Pa
n = 0.0277         # dimensionless
a = 1.7            # dimensionless
E = 5.2e10         # Pa (Elastic modulus)
```

Strain rate sensitivity of stress m

- As $m \rightarrow 0$, rate insensitive (yield)
- As $m \rightarrow 1$, small stress change causes big change in strain rate



Flow rule

Tensorial Flow Rule
(directional form)

$$\mathbf{D}^p = \dot{\epsilon}^p \left(\frac{3}{2} \frac{\mathbf{T}'}{\bar{\sigma}} \right)$$

Equivalent Stress
Definition

$$\sqrt{\frac{3}{2} \mathbf{T}' : \mathbf{T}'}$$

Plastic Strain Rate

$$\dot{\epsilon}^p = A \exp \left(- \frac{Q}{RT} \right)$$

Full Flow Rule

$$\mathbf{D}^p = A \exp \left(- \frac{Q}{RT} \right) \left(\frac{3}{2} \frac{\mathbf{T}'}{\bar{\sigma}} \right)$$

Evolution Equation for the Stress

Stress Evolution Equation
(Rate form of Hooke's Law)

$$\dot{\mathbf{T}} = \mathbb{L} [\mathbf{D} - \mathbf{D}^p] - \mathbf{\Pi} \dot{\theta}$$

(rate-form Hooke's law
for finite deformation
plasticity, with frame

Material Tensors and

- $\mathbb{L} = 2\mu\mathbf{I} + \left(\kappa - \frac{2}{3}\right)$
— isotropic elasticity
- $\mathbb{L}\mathbf{D}$ represents how
instantaneous strain
generate stresses a
to the elastic materi
stiffness properties.

Stress Evolution and Thermal Effects

Stress Evolution and Thermal Effects

In the stress evolution equation,

$$\dot{\mathbf{T}} = \mathbb{L} [\mathbf{D} - \mathbf{D}^p] - \mathbf{\Pi} \dot{\theta},$$

the term $\mathbf{\Pi} \dot{\theta}$ represents the stress change that would occur due to pure thermal expansion

Why Subtract the Thermal Term?

- Thermal expansion creates strain even without external forces.

Relaxed (Intermediate) Configuration

Context for the
Relaxed
Configuration

- The relaxed configuration represents the material after removing plastic

What Happens in the
Relaxed
Configuration?

- The elastic deformation gradient F^e is measured from the relaxed state

Relaxed Configuration Constitutive Laws

Kinematics in the Relaxed Configuration

- Elastic deformation gradient:

$$F = F^e F^p \quad \Rightarrow \quad F^e = F F^{p-1}$$

- Elastic right Cauchy-Green tensor:

$$C^e = F^{eT} F^e$$

Stress

- Kirchhoff (stress):

- Stress p

Dissipation Separation: Elastic vs Plastic in Anand's Model

Thermodynamic Separation

1. **Start with Total Dissipation:**

$$\mathcal{D} = \dot{\omega} - \dot{\psi} \geq 0$$

where

Key Physical Insights

- **Elastic deformations** are recoverable and do not contribute to permanent dissipation.

Reference Configuration

Framework in the Reference Configuration

- The free energy ψ is defined relative to the reference configuration.
- State

Key Equations in the Reference Configuration

- Free energy:

$$\psi = \psi(E^e, \theta, \bar{g}, \bar{\mathbf{B}}, \dots)$$

- Dissipation inequality:

$$\dot{\psi} + \eta \dot{\theta} - \rho_0^{-1} \mathbf{S} : \dot{\mathbf{E}} + (\rho_0 \theta)^{-1} \dot{\rho}_0 \theta \geq 0$$

- Constitutive relation:

Thermodynamics

Thermodynamic Quantities

- Free energy density:

$$\psi = \epsilon - \theta \eta$$

- Reduced dissipation inequality:

$$\dot{\psi} + \eta \dot{\theta} - \rho^{-1} \mathbf{T} : \mathbf{L} + (\rho \theta)^{-1} \mathbf{q} \cdot \mathbf{g} \leq 0$$

- State variables:

• S

• W